Final Project Report — Programming Fundamentals

University Name: **University of Computer & Emerging Sciences Karachi Campus**

Department: **Department of Computer Science**

Course: **Programming Fundamentals**

Project Title: **Supermarket Management System**

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Abstract

The supermarket management system is console-based C program. The main goal was to create a simple, dual-sided tool to help streamline everything from stocking shelves to ringing up a customer. The system is designed with two distinct roles, One is **Admin** module which acts handle back side inventory management which act as staff of the mart which perform several tasks like adding product to the stock ,updating the product ,searching for specific product ,deleting product , Displaying of sale report and handle the low stock mechanism ,Second is **Customer** module which provides a smooth shopping experience. Customer can easily browse the current stock and build their shopping cart and generate the final bill as bonus when customer add more than limit purchase then also apply discount.

1. Introduction

Managing Customer records manually is time-consuming and error-prone. Super market often need a simple digital solution to handle these tasks quickly and accurately. This project provides a basic computerized system where customer can enter desirable product to cat for shopping and instantly generate total bill. It strengthens understanding of variables, conditionals, loops, and modular programming techniques.

2. Objectives

- To Implement Robust Inventory Control.
- To Generate Accurate Transactional Bills and Reports.
- To display bill and sale report in a structured format.
- To reinforce programming concepts like loops, arrays, and functions.
- To provide a simple, user-friendly menu-based interface.

3. System Design

System Overview

Flow of the program:

A. Main System Entry Flow

Start ---→Display Main Menu --→Select (Admin or User) ----→(Proceed to specific module) ---→Exit

B. Administrative Inventory Cycle

Admin Login -→Authenticate Password -→Display Admin Menu -→ Basic CRUD Operation(add, search ,delete ,update)-→Modify Product -→Save Updated Inventory --→View Sales Report -→Return to Main Menu

C. Customer Transaction Cycle

User Section -→ Input Customer Details -→ Display Available Products --→Add Item to Cart (Check Stock) -→ Generate Bill -→ Calculate Total & Apply 5% Discount -→ Update Stock --→ update Record Sale-→Return to Main Menu.

Algorithm

1. Start
2. Initialize product & admin files
3. Show Main Menu
4. If **Admin**
5. Verify password
6. Admin Menu:
    a. Add Product
    b. View Products
    c. Update Product
    d. Delete Product

    e. Search Product
    f. View Sales Report
7. If **User**
8. Enter Name & Contact
9. User Menu:
    a. View Products
    b. Add to Cart
    c. Generate Bill → Apply discount → Save sales → Update stock → Clear cart
10. If Exit → Stop Program

Input & Output

Input: username ,contact number, admin password ,product ID ,product quantity , Product name(search & adding new product).

Output: sale report, product for view, Total bill

4. Implementations

This project is implemented in the C programming language using core concepts such as file handling structure arrays loops conditionals and modular functions the system uses text files to store persistent data for product admin credentials sales history and customers cart.

Language: C

Compiler/IDE: Code::Blocks / Dev C++ / GCC

Key Features

- Persistent storage using text files
- stock management with automatic low stock warning
- unique or repeated product ID handling
- Cart merging (if item already exists > quantity updates)

- customers contact number validation(exactly 11 numbers)
- Automatic discount system
- clean, structured billing display
- Multi menu navigation system

Code Snippet

```
float calculateBill(struct Product cart[], int cartSize) {
   float total = 0;
   for (int i = 0; i < cartSize; i++) {
      float itemTotal = cart[i].price * cart[i].quantity;
      total += itemTotal;
   }return total;}
```

Sample Output

```
WELCOME TO STORE MANAGEMENT SYSTEM

1. Admin Login
2. Customer
Enter your choice: 2

----- Available Products -----
iD   name        price   stock
1    Sugar        80       15
2    Rice        160       20
3    Oil         450       10

Enter product id to added to cart: 2
Enter Quantity: 3
Added successfully
Enter product id for add to cart: 3
Enter quantity: 2
Added successfully
```

```
Add more? (1/0):  zero(0)

------ BILLING ------
product      quatity     price     total
Rice     ---------2----------160  ----480
Oil       --------- 2 --------- 450 --- 900


-----------------------------
grand total:          1380
```

5. Testing & Results

Test No	Input	Expected Output	Acual Output	Status
1	Add product with existing ID	Show Overwrite/increase option	Works correctly	✔
2	Buy product with quantity>stock	Show Not enough stock	Works correctly	✔
3	Generate bill with empy cart	Show cart is empy	Works correctly	✔
4	Admin wrong password	Incorrect password	Works correctly	✔
5	Contact number 10 digit	Invalid contact	Correct Validation	✔
6	After buying product	Stock must decrease	Stock updated in product	✔

The program used successfully for all test and is accurate for all the tests.

5.Conclusion, Limitations & References

Conclusion

Supermarket management system provide a complete solution of managing inventory,purchasing ,and sales record.

- File handling
- Arrays and structure
- Menu-driven programming
- Data Validation

The system ensures accuracy, remove calculations errors and provide a smooth experience for admin and customer.

Limitations

- Console based(Not GUI BASED)
- Not database file used
- No Login systems for multiple systems
- Search option through Id

Future Enhancements

- Login system for many users in future
- Databse will be used in future
- Make the categories of product

- Search product through names
- Online Order module will be added

References

- Complete C programming by College Wallah
- https://www.w3schools.com/c/